ALUMINIUM INGOT MANUFACTURING

Operation, Maintenance & Troubleshooting Manual

Model Series: AIM-5000 / AIM-7500 / AIM-10000

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WARNING: Read all safety instructions before operating or maintaining any equipment described in this manual. Failure to comply may result in serious injury, death, or equipment damage.

TABLE OF CONTENTS

SECTION 1 — GENERAL DESCRIPTION & SPECIFICATIONS

The AIM-Series aluminium ingot manufacturing system is a fully integrated production line designed for the continuous casting of primary and secondary aluminium alloys. The system comprises a charge pre-processing stage, a reverberatory/tilting rotary furnace, an inline degassing unit, a launder transfer system, and a DC (direct-chill) casting machine capable of producing vertical semi-continuous ingots (VDC) or horizontal continuous cast (HCC) rod and slab.

This manual covers the AIM-5000 (5 t/hr), AIM-7500 (7.5 t/hr), and AIM-10000 (10 t/hr) capacity variants. Differences between variants are noted where applicable.

1.1 System Configuration

A standard AIM installation consists of the following major sub-systems:

- **Charge Preparation Bay:** Scrap sorting, baling, shredding, decoating (rotary drum), and pre-heat conveyor.
- **Melting Furnace:** Natural-gas or dual-fuel fired reverberatory furnace (50–100 t capacity) with oxygen enrichment option.
- **Holding/Transfer Furnace:** Electric resistance or gas-fired tilt furnace for alloying additions and temperature homogenisation.
- **Inline Degasser (ILD):** Rotating impeller rotor for hydrogen removal; optional ceramic foam filter (CFF) unit downstream.
- **Casting Pit:** DC casting table, water-cooling ring, starter blocks, level controller, and hydraulic drive.
- **Homogenisation Furnace:** Walking-beam batch furnace for heat treatment of freshly cast ingots.

1.2 Technical Specifications

Parameter	AIM-5000	AIM-7500	AIM-10000	Unit
Melting Capacity	5.0	7.5	10.0	t/hr
Furnace Shell Volume	50	75	100	tonnes Al
Operating Temperature (Furnace)	700–780	700–780	700–780	°C
Melt Temperature (Cast)	690–720	690–720	690–720	°C
Casting Speed (VDC)	60–120	60–120	60–120	mm/min
Ingot Dimensions (Slab)	up to 600×1650	up to 600×1650	up to 750×2000	mm
Ingot Weight (T-bar)	15–22	15–22	20–28	kg/pc
Hydrogen Level (after ILD)	≤0.10	≤0.10	≤0.10	ml/100g Al
Fuel Consumption (Melting)	≤650	≤650	≤640	kCal/kg Al
Installed Electrical Load	480	680	900	kVA
Cooling Water Flow (Casting)	1200–1600	1500–2000	2000–2800	l/min
Cooling Water Pressure	1.5–2.5	1.5–2.5	1.5–2.5	bar

NOTE: All specifications are at standard operating conditions. Contact Process Engineering for non-standard alloy programmes or modified product dimensions.

1.3 Alloy Capability

Alloy Family	Common Designations	Primary Application	Max Si (%)
1xxx Pure Al	1050, 1060, 1100	Electrical conductors, foil	0.25–0.45

Alloy Family	Common Designations	Primary Application	Max Si (%)
3xxx Al-Mn	3003, 3004, 3105	Building products, can sheet	0.30 max
5xxx Al-Mg	5052, 5083, 5182	Marine, automotive, packaging	0.25 max
6xxx Al-Mg-Si	6060, 6061, 6063	Extrusion billet, structural	0.40–0.80
8xxx Special	8006, 8011	Foil, finstock	0.50–1.0

SECTION 2 — FURNACE OPERATIONS

The reverberatory melting furnace is the thermal heart of the AIM system. Correct furnace operation directly determines metal quality, energy efficiency, and refractory service life. Operators must be trained and certified before unsupervised furnace operation.

WARNING: Molten aluminium at 700–780 °C presents extreme burn and explosion hazards. ALL personnel in the furnace bay must wear full-face shield, heat-resistant gloves (rated ≥1000 °C), aluminised spats, and flame-retardant coverall. Wet tools or charge materials introduced into the melt WILL cause a violent steam explosion.

2.1 Start-Up Procedure

Perform the following checks before every melting campaign:

Step	Action	Accept Criterion	Responsible
1	Inspect furnace shell and door seals for cracks or erosion	No cracks >5 mm; seals intact	Furnace Operator
2	Check burner pilot ignition and flame scanner signal	Pilot lights within 5 s; scanner OK	Furnace Operator
3	Verify combustion air fan operation and pressure	Fan running; duct pressure 15–25 mbar	Furnace Operator
4	Confirm natural gas supply pressure at header	100–140 mbar (g)	Utilities Technician
5	Inspect charging door mechanism — manual override tested	Door opens/closes freely	Furnace Operator
6	Check thermocouple Type-K readings on DCS screen	All TCs reading; no "OPEN" fault	Control Operator
7	Confirm flue gas damper position (start: 30% open)	Damper at 30±5%	Furnace Operator
8	Record pre-heat metal temperature if heel present	Log in batch record	Furnace Operator
9	Enable combustion control loop on DCS	Loop in AUTO; setpoint confirmed	Control Operator
10	Insert emission probe for initial O2/CO baseline	O2: 2–4%; CO <100 ppm	Environment Tech

2.2 Operating Temperature Zones

Zone	Location	Set Temp (°C)	Trip Temp (°C)	Function
Zone 1 — Charging	Charging end / door area	700–730	760	Melt incoming scrap charge
Zone 2 — Melting	Central bath	730–760	790	Complete melting; alloying
Zone 3 — Tapping	Tapping/skimming end	710–730	760	Temp. homogenisation before tap
Crown (roof)	Furnace crown	800–860	920	Radiant heat transfer
Flue Gas Exit	Stack entry	350–500	600	Combustion efficiency monitoring

WARNING: If any zone temperature exceeds its TRIP TEMPERATURE, the furnace control system will shut off all burners automatically. Do NOT manually override this safety interlock. Investigate and resolve the root cause before restarting.

2.3 Charging Procedure

CAUTION: Never charge loose coiled wire, hollow castings, or sealed canisters directly into the melt. Trapped moisture or air causes explosive steam/pressure release. All charge must pass moisture check (<0.1% surface moisture).

Charging sequence for a 50-tonne heat (reference alloy 6063):

Order	Material	Quantity (kg)	Condition Requirement
1 (Heel)	Molten aluminium heel (prior heat)	8,000–12,000	Temp ≥680 °C; clean surface
2	Clean 6063 extrusion scrap (class A)	18,000–22,000	Pre-heated ≥200 °C; dry
3	Foundry returns / re-melts	5,000–8,000	Pre-heated; screened for inclusions
4	Primary aluminium ingots (99.7%)	Balance to heat weight	Dry; stacked, not thrown
5	Alloying master alloys (see §5)	Per chemistry target	Added after full melt; see §5
6	Fluxing salt (if required)	3–5 kg/t metal	After melt-down; see §5.3

2.4 Furnace Performance Monitoring

Parameter	Method	Frequency	Target / Limit
Bath temperature	Immersion thermocouple (Type-K)	Every 30 min	700–760 °C
Combustion excess air	Flue gas analyser (O2)	Continuous	O2: 2–4%
CO emissions	Flue gas analyser	Continuous	<200 ppm (alarm at 150 ppm)
Fuel consumption	Mass flow meter	Per heat	≤650 kCal/kg Al melted
Dross generation	Gravimetric (weighed skip)	Per skim	<3.5% of melt weight
Tap-to-tap time	DCS batch timer	Per heat	AIM-5000: 90–110 min; AIM-7500: 110–130 min
Metal loss	Charge weight vs tap weight	Per heat	<2.0%
Refractory surface temp	Infrared pyrometer	Weekly	<65 °C external shell

SECTION 3 — CASTING OPERATIONS

The DC casting pit converts molten aluminium from the holding furnace into semi-fabricated ingots via direct-chill solidification. The casting pit is classified as a Restricted Access Zone. Entry during an active cast is permitted only for essential operational tasks and requires a signed Permit to Work.

WARNING: Water contact with molten aluminium causes an instantaneous violent steam explosion. Ensure all casting pit surfaces, tooling, launder liners, and starter blocks are DRY and pre-heated before introducing molten metal. Minimum pre-heat: 250 °C for launder; 200 °C for starter block surface.

3.1 Pre-Cast Checklist

Item	Check	Accept Criterion	Sign-Off
Launder system	Visual inspect for cracks, spalling, moisture	No defects; surface temp ≥250 °C	Cast Supervisor
Metal level sensor	Float/laser level calibration	±2 mm accuracy vs manual gauge	Control Operator
Starter block position	Hydraulic table at "start" height	Block surface 50 mm below mould lip	Cast Operator
Water flow — primary	Verify flow rate and pressure	Flow per Table 3-A; 1.5–2.5 bar	Cast Operator
Water flow — secondary	Check sump drain clear; no standing water	Sump dry; drain valve open	Cast Operator
Mould oil feed	Verify lubrication pump running	0.3–0.5 l/hr per mould	Cast Operator
Casting speed drive	Test jog function both directions	Table moves smoothly; no alarms	Cast Operator
Emergency dump valve	Test manual actuation	Launder drains within 15 s	Cast Supervisor
Gas purge (launder)	Confirm N2 purge flow active	N2 flow 2–4 Nm3/hr	Control Operator
Metal temperature	Thermocouple at tundish entry	690–710 °C (see Table 3-B)	Shift Metallurgist

3.2 Casting Temperature by Alloy — Table 3-B

Alloy	Cast Temp (°C)	Speed (mm/min)	Water Flow (l/min/mould)	Notes
1050/1060	685–695	100–120	180–220	Low superheat; rapid solidification
3003	695–705	90–110	160–200	Mn addition sensitive to temp
5052	700–715	80–100	150–190	Mg oxidation — inert gas cover required
5083	705–720	70–90	140–180	High Mg; strict temp control ±5 °C
6060	695–710	85–105	155–195	Standard billet programme
6061	700–715	80–100	150–190	Mg+Si balance critical
6063	690–705	90–110	160–200	Most common extrusion billet
8011	695–710	95–115	165–205	Fe/Si ratio ≤0.1 required

CAUTION: Casting temperature must not exceed the upper limit by more than 5 °C. Excessive superheat increases hydrogen pick-up, porosity, and hot-tearing susceptibility.

3.3 Start-Up Sequence (VDC Cast)

Step	Action	Time Ref
1	Confirm all pre-cast checklist items completed and signed	T–30 min
2	Start mould oil lubrication pump; confirm flow at all moulds	T–20 min
3	Start primary cooling water at low flow (30% of target)	T–15 min
4	Begin launder pre-heat; confirm launder temp ≥250 °C	T–10 min
5	Open holding furnace tap; fill launder to 60% level	T–5 min
6	Increase cooling water to 60% of target flow rate	T–3 min
7	Fill tundish; metal level sensor activated	T–1 min
8	Metal fills mould; wait for steady meniscus (no splashing)	T=0 cast start
9	Start hydraulic table descent at low speed (20 mm/min)	T+30 s
10	Ramp casting speed to target over 3–5 minutes	T+3 to T+5 min
11	Increase primary cooling water to 100% target flow	T+5 min
12	Log all parameters on casting record form AIM-F-031	T+10 min

3.4 Ingot Quality Parameters

Quality Attribute	Measurement Method	Specification	Reject Criterion
Surface quality	Visual / profilometer	Ra ≤6.3 µm; no cold shuts, bleeds, laps	Any cold shut or bleed
Dimensional tolerance (width)	Vernier caliper / laser gauge	Nominal ±3 mm	>±5 mm
Dimensional tolerance (thickness)	Vernier caliper	Nominal ±2 mm	>±4 mm
Porosity (X-ray)	Real-time X-ray scanner	≤0.5 mm isolated pores; no clusters	Cluster porosity
Hydrogen content	Alspek/Telegas (in-line)	≤0.10 ml/100g Al	>0.15 ml/100g Al
Chemical composition	OES spectrometer (per heat)	Within EN573/ASTM B209 limits	Any element out of spec
Macrostructure (etch)	Baumann print / macro etch	Equiaxed grain; no centreline cracking	Any cracking
Ingot butt shape	Visual / template gauge	Butt swell ≤8 mm	>12 mm swell

SECTION 4 — PRE-PROCESSING & CHARGE PREPARATION

Charge quality is the single most controllable factor in determining furnace efficiency and final metal quality. Contaminated or insufficiently prepared charge leads to increased dross, elevated hydrogen, poor chemistry control, and reduced refractory life.

4.1 Scrap Classification

Class	Description	Max Contaminants	Pre-Treatment
Class A — Clean Process Scrap	In-house gates, runners, trimmings	None	None; direct charge
Class B — Clean Purchased Scrap	New extrusion profiles, sheet offcuts	<0.5% oil/moisture	Visual check; pre-heat if damp
Class C — Painted / Coated Scrap	Painted profiles, lacquered can scrap	<3% coating by weight	Mandatory decoating in rotary drum
Class D — Mixed Turnings / Borings	Machining chips, swarf	<5% coolant oil	Dewatering, briquetting, pre-heat
Class E — Dross / Skimmings	Dross ball, black dross, skimmings	Variable	Dross press; composition analysis
Class F — Contaminated / Unknown	Mixed or unknown origin scrap	Unknown	Spectro analysis; quarantine until cleared

WARNING: Class F scrap must never be charged without written approval from the Shift Metallurgist and a full OES composition check. Unknown alloy families may contain Be, Pb, Cd, or other toxic elements that create hazardous fumes at melt temperatures.

4.2 Decoating Rotary Drum — Operating Parameters

Parameter	Set Point	Alarm Limit	Trip Limit
Drum inlet temperature	450–480 °C	500 °C	520 °C
Drum outlet gas temperature	350–400 °C	420 °C	440 °C
Drum rotation speed	3–5 rpm	—	>8 rpm
Afterburner temperature	800–850 °C	780 °C (low)	900 °C (high)
Residence time	15–25 min	—	—
Throughput (Class C scrap)	2–4 t/hr	—	—
Organic removal efficiency	≥95% coating removal	<90%	<85% (stop; inspect)
Exhaust SO2+HCl (combined)	<50 mg/Nm3	40 mg/Nm3	50 mg/Nm3

4.3 Moisture Control

Material Type	Max Moisture	Test Method	Frequency
All scrap (classes A–D)	0.10% by weight	Moisture meter or oven loss	Per lot; after rainfall
Primary ingots (from outdoor storage)	0.05% (visually dry)	Visual; wipe test	If stored outdoors
Alloying master alloys	0.05%	Visual; moisture meter	Each drum opened
Fluxing salts (loose)	0.20%	Oven loss method	Each bag/drum opened

Material Type	Max Moisture	Test Method	Frequency
Launder / trough liners	Surface temp ≥250 °C	Contact thermometer	Before each use

CAUTION: Charge materials that feel damp, show condensation, or have been rained on must be pre-dried at 200–250 °C in the pre-heat conveyor oven (minimum 45 minutes) before furnace loading. Do not attempt to "dry" material by placing it near the furnace door.

SECTION 5 — ALLOYING & DEGASSING

5.1 Alloying Additions

Alloying elements are added to the holding/transfer furnace after melt-down is complete and the bath is at 730–750 °C. All additions must be logged in the batch record (Form AIM-F-015) and confirmed by the Shift Metallurgist.

Element	Addition Form	Add Temp (°C)	Recovery	Notes
Silicon (Si)	AlSi20 master alloy or pure Si grain	730–750	0.90–0.95	Stir well after addition; Si segregates if not mixed
Manganese (Mn)	AlMn20 master alloy	740–760	0.85–0.92	Requires vigorous stirring; slow dissolution
Magnesium (Mg)	Pure Mg ingot or AlMg25	700–730	0.80–0.90	Add last; Mg oxidises at high temp; use plunger or bell
Copper (Cu)	AlCu50 master alloy or pure Cu	740–760	0.92–0.98	Fast dissolution; avoid CuO contamination
Iron (Fe)	AlFe20 master alloy (correction only)	740–760	0.90–0.95	Only for correction; rarely added intentionally
Titanium (Ti)	AlTi5B1 rod (grain refiner)	690–710	—	Add inline to launder only — NOT to furnace bath
Strontium (Sr)	AlSr10 rod (modifier)	690–710	—	Add inline or as waffle; sensitive to holding time
Boron (B)	AlB3 master alloy	730–750	0.80–0.88	For 1xxx conductor grade; reduces conductivity loss

WARNING: Magnesium metal reacts violently with moisture and with chlorine-based fluxes. Never add Mg to a bath that has just received salt flux. Wait minimum 10 minutes and stir the bath thoroughly before Mg addition.

5.2 Inline Degassing Unit (ILD)

Parameter	Specification	Unit
Rotor speed	400–600	rpm
Process gas	N2 (standard) or Ar/Cl2 blend (95/5)	—
Gas flow rate	8–15	Nl/min
Metal residence time	4–8	min
Target H2 level (exit)	≤0.10	ml/100g Al
Degassing efficiency	≥50% H2 reduction	—
Operating temperature	700–730	°C
Graphite rotor inspection interval	Every 500 t processed	tonnes
Rotor replacement threshold	Erosion >15% of original diameter	—

CAUTION: Never use Cl2-containing gas mixtures without confirmed operation of the fume extraction system and valid gas detector calibration. Chlorine gas is acutely toxic; TLV-C = 1 ppm.

5.3 Salt Flux Application

Flux Type	Composition	Rate (kg/t)	Purpose	When Applied
Cover flux	60% NaCl / 40% KCl	3–5	Dross separation; metal protection from oxidation	After melt-down; before alloying
Refining flux	50% NaCl / 30% KCl / 20% NaF	2–4	Inclusion and Na removal (esp. 5xxx)	Before tapping; stir 10 min
Drossing flux	70% NaCl / 30% KCl	5–8 (on dross)	Dross consolidation; metal recovery improvement	Sprinkled on dross before skimming

NOTE: Flux additions generate salt fume and HCl/HF vapour. Always operate the furnace fume extraction system at maximum capacity during and after flux treatment. Fume extraction failure is an immediate work-stop condition.

SECTION 6 — MAINTENANCE SCHEDULE

All maintenance must be performed by trained personnel and documented in the plant CMMS. Planned maintenance windows are co-ordinated with the Production Planner. Before any maintenance on the furnace or casting pit, a Lockout/Tagout (LOTO) permit must be issued and all energy sources isolated.

WARNING: LOCKOUT/TAGOUT (LOTO): Before any maintenance work on the furnace, casting pit, or fluid systems, obtain and follow the equipment-specific LOTO procedure. Do not rely on process interlocks alone. All energy sources (gas, electric, hydraulic, pneumatic, thermal) must be isolated and tagged.

6.1 Daily / Per-Shift Maintenance

Task	Equipment	Action	Accept Criterion
Check burner flame	Melting furnace	Visual through sight glass	Stable blue/orange flame; no blow-off
Inspect launder joints	Launder system	Visual; check for metal seepage	No seepage; joints sealed
Check ILD rotor condition	Inline degasser	Visual — check erosion, fracture	No visible cracks or major chips
Check cooling water flow	Casting pit	Verify all flow indicators reading	All flows within ±10% of target
Lubricate casting table chain	Casting table	Apply food-grade grease to chain	Even coverage; no dry links
Drain condensate from gas lines	Furnace gas supply	Open drain valves on moisture traps	Water discharged; traps clear
Check thermocouple readings	All furnace zones	Compare DCS vs handheld IR pyrometer	±15 °C agreement
Skim dross from holding furnace	Holding furnace	Use clean skimmer; collect in dross skip	Metal surface clean before tap

6.2 Weekly Maintenance

Task	Equipment	Duration	Personnel
Refractory inspection — full visual	Melting furnace	2 hr (cold)	Refractory Specialist
Burner nozzle cleaning and gap check	All burners	30 min/burner	Burner Technician
Mould face inspection and dressing	Casting moulds	1 hr/mould set	Cast Technician
Hydraulic oil level and filter check	Casting table hydraulics	30 min	Mechanical Technician
ILD graphite rotor measurement	Inline degasser	45 min	Cast Technician
Calibrate metal level sensors	Casting tundish	1 hr	Instrument Technician
Check and clean gas detector heads	Casting pit and furnace bay	2 hr	Safety Officer
Inspect and test emergency dump valve	Launder dump system	30 min	Cast Supervisor
Grease casting table guide columns	Casting table	1 hr	Mechanical Technician
Inspect decoating drum trunnion bearings	Decoating drum	45 min	Mechanical Technician

6.3 Monthly / Shutdown Maintenance

Task	Trigger / Interval	Action	Sign-Off
Full refractory repair	Monthly or wear >30 mm	Patch guniting or full reline per plan	Refractory Specialist + Plant Manager

Task	Trigger / Interval	Action	Sign-Off
Thermocouple replacement (all zones)	Every 3 months or on failure	Replace Type-K tips; recalibrate DCS	Instrument Technician
Combustion system overhaul	Every 6 months	Strip/inspect/replace burner bodies, pilots, scanners	Burner OEM Technician
Casting table hydraulic oil change	Every 6 months or 2000 hr	Drain, flush, refill with ISO VG46	Mechanical Technician
ILD graphite rotor replacement	Every 500 t or >15% diameter loss	Replace rotor assembly; inspect shaft seal	Cast Supervisor
Complete mould set replacement	Every 80–120 casts or on defect	Replace mould faces, o-rings, water connections	Cast Supervisor
Decoating drum internal inspection	Every 6 months	Enter vessel (confined space permit required)	Safety Officer + Mech Tech
Full launder reline	Annually or on cracking	Remove and relay launder refractory castable	Refractory Specialist

SECTION 7 — TROUBLESHOOTING GUIDE

This section provides systematic fault identification and corrective action for the most common problems encountered in aluminium ingot manufacturing. Refer to Section 8 for DCS/PLC fault code definitions.

> **NOTE:** The troubleshooting tables follow the format: PROBLEM → PROBABLE CAUSES → CORRECTIVE ACTIONS → WARNING/CAUTION. Always complete a Root Cause Analysis (Form AIM-F-042) for any fault that causes production downtime exceeding 30 minutes.

7.1 Furnace — High Metal Hydrogen (H_2 > 0.15 ml/100g Al)

PROBLEM	Metal hydrogen content exceeds 0.15 ml/100g Al after ILD treatment; casting quality rejection risk.
PROBABLE CAUSES	1. Wet or damp charge materials introduced to furnace 2. Worn or cracked ILD graphite rotor — inefficient bubble dispersion 3. ILD gas flow too low or gas supply pressure drop 4. Excessive furnace atmosphere humidity (ambient RH >80%) 5. Contaminated process gas (N_2 purity <99.5%) 6. Holding furnace cover gas (N_2) flow interrupted 7. Metal temperature too high (>750 °C) — increased H_2 solubility 8. Furnace refractory saturated with moisture (after cold start)
CORRECTIVE ACTIONS	1. Quarantine current charge lot; test moisture content — reject if >0.1%. 2. Stop ILD; measure rotor diameter. If erosion >15%, replace rotor before restart. 3. Check N_2 supply pressure at ILD inlet header (3–5 bar). Verify rotameter at 8–15 Nl/min. 4. Review plant humidity log. Extend ILD degassing time or increase gas flow to 15 Nl/min. 5. Request N_2 purity certificate from supplier; switch to backup cylinder bundle if suspect. 6. Restore cover gas flow; check solenoid valve on N_2 purge line. Flush metal surface. 7. Reduce holding furnace setpoint 15–20 °C; allow 20 min equilibration before re-sampling. 8. Extend bake-out cycle on restart; minimum 8 hr at 600 °C before first charge.
WARNING	Do not cast metal with H_2 >0.20 ml/100g Al. Ingots will exhibit subsurface porosity and blister during downstream processing. Divert melt to scrap via the emergency dump if H_2 cannot be reduced within one additional ILD pass.

7.2 Casting — Cold Shut / Misrun on Ingot Surface

PROBLEM	Horizontal fold lines (cold shuts) or incomplete fill areas (misruns) visible on ingot surface after cast.
PROBABLE CAUSES	1. Casting temperature too low (metal below liquidus + superheat) 2. Metal level fluctuation in mould — unstable float/level controller 3. Insufficient or blocked mould lubrication (oil pump fault) 4. Casting speed too high for given metal temperature 5. Interrupted metal flow from holding furnace (tap blockage, slag build-up) 6. Mould water flow too high — premature skin freeze
CORRECTIVE ACTIONS	1. Increase holding furnace setpoint by 10 °C; verify with immersion thermocouple before restarting. 2. Check float arm for damage or hang-up; verify laser level sensor clean and calibrated. 3. Check mould oil pump pressure (0.3–0.8 bar); inspect manifold for blockages; prime pump. 4. Reduce casting speed by 10 mm/min increments; target speed in Table 3-B. 5. Rod or replace furnace tap; check launder slope and remove any slag bridges. 6. Reduce mould water flow to lower limit in Table 3-B; increase by 5% if butt swell occurs.
WARNING	Cold shuts are a serious structural defect. Mark affected length with red paint and quarantine ingots for metallurgical disposition. Do not ship to customer without written release from Quality Manager.

7.3 Furnace — Excessive Dross Generation (> 3.5% of Melt Weight)

PROBLEM	**Dross weight on skim exceeds 3.5% of melt weight; increased metal loss and operating cost.**
PROBABLE CAUSES	1. Charge materials heavily contaminated with paint, organics, or moisture 2. Furnace bath over-temperature (>770 °C) accelerating oxidation 3. Aggressive mechanical stirring causing turbulence and air entrainment 4. Excessive dwell time between charge and skim (>60 min without cover flux) 5. Combustion excess air too high (O_2 >6%) — oxidising atmosphere 6. Insufficient or incorrect flux addition 7. High Mg alloy without adequate inert cover gas
CORRECTIVE ACTIONS	1. Audit charge quality; reject or re-route Class C/D scrap for additional decoating. 2. Reduce Zone 2 setpoint by 15 °C; confirm bath thermocouple accuracy. 3. Reduce stirrer speed or paddle traverse rate; use gentle figure-8 pattern. 4. Apply 3 kg/t cover flux within 10 min of melt-down; skim within 45 min. 5. Reduce combustion air damper opening; target flue O_2: 2–4%. 6. Verify flux type and rate per Section 5.3; consider adding refining flux. 7. Increase N_2 cover gas flow over bath; check curtain seal on furnace door.
CAUTION	Dross recovery by dross press is only effective on freshly-skimmed hot dross. Cooled dross loses metal recovery efficiency. Transfer dross skip to dross press within 15 minutes of skimming.

7.4 Casting — Ingot Butt Crack (Vertical Crack in Butt Section)

PROBLEM	**Vertical or radial cracks appear in the butt (bottom) section of the ingot shortly after cast start.**
PROBABLE CAUSES	1. Casting speed ramped too quickly from start speed to target speed 2. Cooling water applied at full flow from cast start (thermal shock to butt) 3. Starter block surface temperature too cold (<100 °C) 4. Metal temperature too high — excessive sump depth during start-up 5. Alloy highly susceptible to hot tearing (e.g. 2xxx, 7xxx, some 5xxx) 6. Incorrect starter block profile — poor fit to mould bottom
CORRECTIVE ACTIONS	1. Follow speed ramp in Section 3.3: 20 mm/min for first 30 s; ramp over 3–5 min to target. 2. Apply water at 30% flow for first 3 min; ramp to 60% at T+3 min; 100% at T+5 min. 3. Pre-heat starter block to 200 °C minimum; verify with contact thermometer. 4. Reduce holding furnace setpoint; target metal temp at lower end of Table 3-B range. 5. Consult alloy-specific casting parameters; consider water reduction and slower start speed. 6. Verify starter block drawing number matches mould format; replace if worn or damaged.
WARNING	Butt cracks in the mould can cause an uncontrolled runout of molten metal into the casting pit. If cracks are observed on DCS camera during cast, immediately reduce casting speed to 10 mm/min and call the Cast Supervisor. Do not stop the cast abruptly — frozen ingot in mould is also a serious hazard.

7.5 Chemistry — Out-of-Specification Silicon Content

PROBLEM	**OES analysis shows Si content outside specification range (too high or too low).**
CAUSES — Si TOO HIGH	1. Incorrect master alloy addition calculation (AlSi20 over-added) 2. Higher-than-expected Si in scrap charge (class mismatch or contamination) 3. Previous heat residue in heel — Si carryover not accounted for in calculation
CAUSES — Si TOO LOW	1. AlSi20 master alloy not fully dissolved before sampling 2. Master alloy addition under-calculated due to recovery factor error 3. OES sampling error — sample not representative (surface oxide included)
ACTIONS — Si TOO HIGH	1. Do NOT add more scrap or master alloy. Calculate dilution volume with pure Al ingots. 2. If dilution is not practical (furnace at capacity), discuss with Shift Metallurgist — may blend cast or downgrade to 4xxx. 3. Review scrap charge certificates; update scrap Si database entry.

PROBLEM	OES analysis shows Si content outside specification range (too high or too low).
ACTIONS — Si TOO LOW	1. Wait 5 min after AlSi20 addition; stir bath; re-sample before any corrective addition. 2. Recalculate required addition using recovery factor 0.90–0.95; add balance amount. 3. Re-sample from bath mid-depth (300–400 mm from surface) with clean graphite tube sampler.
CAUTION	Never tap or cast a heat with chemistry outside specification without written sign-off from the Shift Metallurgist. Out-of-spec material cast as in-spec product is a quality fraud and a contractual violation.

7.6 Furnace — Burner Failure to Ignite

PROBLEM	One or more furnace burners fail to ignite or shut off unexpectedly; zone temperature falling.
PROBABLE CAUSES	1. Gas supply pressure below minimum (<80 mbar at burner block) 2. Pilot electrode tip worn or fouled with carbon deposits 3. Flame scanner (UV/IR) — dirty lens or failed detector cell 4. Combustion air fan failure or damper found closed 5. High-limit thermocouple tripped (zone over-temperature event) 6. PLC burner management system (BMS) interlock active — see fault code in DCS
CORRECTIVE ACTIONS	1. Check gas header pressure gauge; contact Utilities if low. Verify manual shutoff valves fully open. 2. Isolate burner (LOTO); remove and clean pilot electrode; set gap to 3–4 mm; replace if worn below 8 mm. 3. Clean scanner lens with dry cloth; use compressed air (not solvent). Test scanner signal on BMS screen. 4. Check air fan rotation direction; reset thermal overload on fan MCC; inspect ducting for blockage. 5. Allow zone to cool below trip setpoint; identify cause of over-temperature; reset BMS after clearance. 6. Read fault code on BMS HMI; refer to Section 8 for code definition and prescribed reset procedure.
WARNING	Never manually force open a gas solenoid valve to try to light a burner that has failed to ignite. Accumulated unburnt gas in the furnace chamber will ignite explosively when the ignition sequence next activates. Always follow the BMS purge-and-restart sequence.

7.7 Casting — Ingot Surface Bleed-Out

PROBLEM	Semi-solid or liquid metal bleeds through solidified ingot shell causing surface tears, fins, or metal runout.
PROBABLE CAUSES	1. Primary cooling water flow too low — inadequate solidified shell thickness 2. Casting speed too high — sump too deep; shell too thin at mould exit 3. Metal temperature at mould too high (above upper limit in Table 3-B) 4. Mould face worn or damaged — uneven water distribution 5. Sudden interruption in cooling water (pump trip or valve failure) 6. Mould oil failure — increased friction causing shell tear on mould exit
CORRECTIVE ACTIONS	1. Increase primary cooling water to upper limit in Table 3-B; check for blocked spray holes in water ring. 2. Reduce casting speed immediately by 15–20 mm/min; stabilise before attempting to re-ramp. 3. Reduce holding furnace temperature; do not cast until metal temp returns to Table 3-B range. 4. Stop cast; inspect mould face for erosion channels; replace mould set if wear score exceeds 2 mm. 5. EMERGENCY: If water loss occurs during active cast, activate emergency dump valve immediately. 6. Check oil pump pressure and flow; unclog distribution manifold; increase oil flow by 20%.
WARNING	Bleed-out of liquid metal into the casting pit water sump will cause a violent steam explosion. If bleed-out is observed, evacuate the casting pit immediately and activate the emergency stop. Do not re-enter the pit until the area is clear, cooled, and declared safe by the Cast Supervisor.

7.8 Pre-processing — Decoating Drum Afterburner High Temperature Trip

PROBLEM	**Afterburner temperature exceeds 900 °C trip limit; drum shuts down; pre-processing production interrupted.**
PROBABLE CAUSES	1. Charge batch contained excessive organic load (heavily painted scrap in oversized lot) 2. Afterburner fuel rate set too high (manual override left active by previous shift) 3. Excess air damper malfunction — insufficient dilution air entering afterburner 4. Thermocouple fault giving false high temperature reading 5. Secondary combustion of accumulated carbonaceous residue in afterburner chamber
CORRECTIVE ACTIONS	1. Stop drum feed immediately. Reduce lot size for heavily coated scrap to 50% of normal charge. 2. Check manual override status on afterburner fuel valve; return to automatic control mode. 3. Inspect and manually open excess air damper; check actuator and positioner calibration. 4. Compare afterburner TC reading with portable infrared pyrometer; replace TC if deviation >30 °C. 5. After cool-down, inspect and clean afterburner refractory; remove carbonaceous deposits.
CAUTION	Do not open the afterburner inspection door while temperature exceeds 200 °C. Allow full cool-down period (minimum 4 hours with induced draft running) before internal inspection. Ensure confined space permit is issued before entry.

SECTION 8 — FAULT CODES REFERENCE

The following DCS/PLC fault codes are generated by the AIM control system. For codes not listed, contact the OEM service line.

Code	System	Description	Category	Immediate Action
F-001	Furnace BMS	Burner 1/2/3 ignition failure after 3 attempts	Alarm / Shutdown	Initiate purge cycle; inspect pilot; see §7.6
F-002	Furnace BMS	Zone 1/2/3 over-temperature trip	Safety Trip	Burners off; investigate; do not override interlock
F-003	Furnace BMS	Flame scanner signal loss	Alarm	Clean scanner; check cable; replace if fault persists
F-004	Furnace BMS	Combustion air fan trip (overload or phase loss)	Shutdown	Check MCC; reset thermal overload; identify cause before restart
F-005	Furnace BMS	Gas pressure low (<80 mbar)	Alarm / Shutdown	Contact Utilities; check manual valves; do not force restart
F-010	ILD Control	ILD rotor drive inverter fault	Alarm	Check drive fault code on inverter display; cycle power after 2 min
F-011	ILD Control	N2 gas flow low (<6 Nl/min)	Alarm	Check N2 supply pressure; inspect rotameter float; clear blockage
F-012	ILD Control	ILD metal temperature high (>740 °C)	Alarm	Reduce holding furnace temp; check thermocouple calibration
F-020	Casting	Casting table drive fault (hydraulic pressure low)	Shutdown	Check hydraulic pump and pressure accumulator; call Mechanical
F-021	Casting	Cooling water flow low (<80% setpoint)	Safety Trip	Stop cast immediately; identify blockage or pump fault; see §7.7
F-022	Casting	Metal level sensor fault (no signal)	Alarm	Switch to manual level control; check float arm or laser unit
F-023	Casting	Mould lubrication oil flow low	Alarm	Check oil pump; inspect manifold for blockage; see §7.2
F-030	Decoating	Afterburner temperature high trip (>900 °C)	Safety Trip	Stop drum feed; see §7.8; do not restart until cause resolved
F-031	Decoating	Drum drive overload trip	Alarm	Check for drum overfill; clear blockage; reset drive MCC
F-040	General	Emergency stop activated (E-stop pressed)	Safety Trip	Identify who activated and why before reset; Supervisor sign-off required
F-041	General	Fume extraction fan fault	Alarm / Work Stop	Cease all fluxing/degassing; evacuate furnace bay; fix before resuming
F-050	Homo. Furnace	Walking beam drive fault	Alarm	Check mechanical jam; inspect pushers; call Mechanical Technician
F-051	Homo. Furnace	Homogenisation furnace over-temperature (>600 °C)	Safety Trip	Burners off; check alloy programme; verify thermocouple accuracy

SECTION 9 — SAFETY & HAZARD SUMMARY

9.1 Hazard Identification Table

Hazard	Location	Risk Level	How to Prevent It
Molten metal contact	Furnace, casting pit, launder	CRITICAL — burns; fatality	Full PPE always; dry tools; no wet charge; trained operators only
Steam explosion	Casting pit, furnace	CRITICAL — explosion; fatality	All surfaces/tools pre-heated ≥250 °C; moisture check on charge; confirm dry pit before cast
Gas explosion (unburnt fuel)	Furnace bay	CRITICAL — explosion; fire	Never bypass BMS purge sequence; follow F-001 procedure; gas detectors functional
Chlorine gas exposure	ILD unit, launder area	HIGH — toxic gas injury	Fume extraction confirmed running; Cl2 detector calibrated before ArCl gas use
Refractory dust (silica/alumina)	Furnace reline, launder work	HIGH — silicosis (long-term)	P3 respirator during refractory work; wet suppression; medical surveillance programme
Fluoride fume (salt flux)	Furnace bay during fluxing	MEDIUM — respiratory/skin	Fume extraction running; NaF-containing flux only with approved engineering controls
Hydraulic fluid injection injury	Casting table hydraulics	HIGH — penetrating injury	Never use hand to check hydraulic leaks; use cardboard; isolate before maintenance
Electrical hazard (HV)	Electrical cubicles, furnace	CRITICAL — electrocution	LOTO mandatory; qualified electrician only; no work on live equipment
Heat stress	Furnace bay, casting pit	MEDIUM — heat exhaustion	Mandatory rest rotation; cool rest area; hydration; buddy system in hot areas
Moving equipment (crane/ladle)	Furnace bay, casting area	HIGH — crush; struck-by	Exclusion zones; audible alarms; spotters; no walking under suspended loads

9.2 Mandatory Personal Protective Equipment (PPE)

Work Area	Minimum PPE Required
Furnace bay — general access	Safety helmet, safety glasses, flame-retardant coverall, steel-toe boots, hearing protection
Near open furnace / tapping	Above PLUS: full-face aluminised shield, aluminised heat-resistant gloves (≥1000 °C rated), aluminised spats/gaiters
Active casting pit	Above PLUS: aluminised apron; face shield sealed; radio communication with control room confirmed before entry
Refractory work	Above PLUS: P3 half-face or full-face respirator; disposable oversuit over coverall
ILD / chlorine gas operation	Full-face supplied-air respirator (SCBA or airline) if Cl2 blend; minimum FF respirator with ABEK filter
Electrical / instrument work	Insulated gloves (Class 1 minimum); face shield for arc flash risk; anti-static coverall
Decoating drum (hot work area)	Heat-resistant gloves; face shield; coverall rated for hot surfaces; steel-toe boots

WARNING: Entry to the casting pit during an active cast is PROHIBITED unless a Permit to Work has been issued, a second person is present at the pit edge, the cast supervisor has been notified, and the emergency stop button location is confirmed by the entrant.

SECTION 10 — WARRANTY & SERVICE

10.1 Warranty Terms

Category	Warranty Period	Coverage
Main structural components (furnace shell, pit frame)	24 months from commissioning	Manufacturing defects; weld failures; dimensional non-conformance
Mechanical rotating equipment (ILD, drum drives, casting table)	12 months from commissioning	Bearing failures; gear defects; shaft failure under normal operating load
Refractory lining (factory-installed)	12 months or first relining, whichever first	Premature spalling or adhesion failure under normal conditions
Electrical / control equipment (BMS, DCS panels)	12 months from commissioning	Component failures under normal operating conditions
Consumable items (graphite rotors, moulds, thermocouple tips)	Not covered — consumables	Life expectancy guidelines in §6; replacement on condition basis
Damage from improper operation, wet charging, or overload	Not covered	Operator error exclusion — refer to §2.3 and §4.3 for correct procedures

10.2 Service Contacts

Service Type	Contact	Hours
Emergency breakdown (safety-critical)	+1-800-AIM-SERV (1-800-246-7378)	24 / 7
Scheduled maintenance / spare parts	service@aimfurnace.com	Mon–Fri 07:00–19:00
Refractory technical support	refrac-support@aimfurnace.com	Mon–Fri 08:00–17:00
DCS/BMS software support	controls@aimfurnace.com	Mon–Fri 08:00–17:00
Spare parts ordering	spares@aimfurnace.com	Mon–Fri 08:00–17:00

NOTE: Always quote your plant name, equipment model (AIM-5000/7500/10000), serial number (on the main panel nameplate), and the DCS fault code when contacting service. This ensures the fastest response and correct parts dispatch.

END OF DOCUMENT